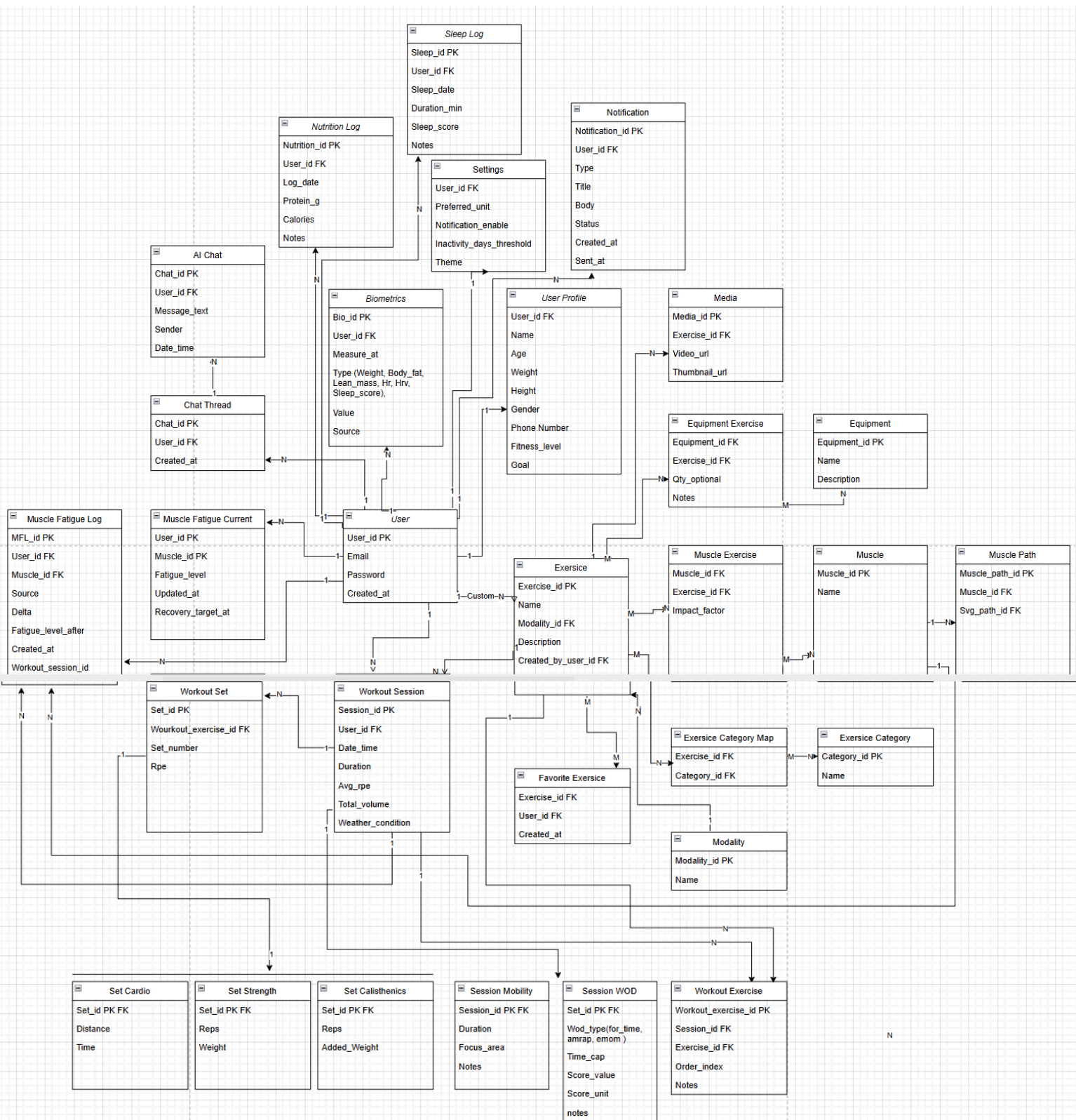
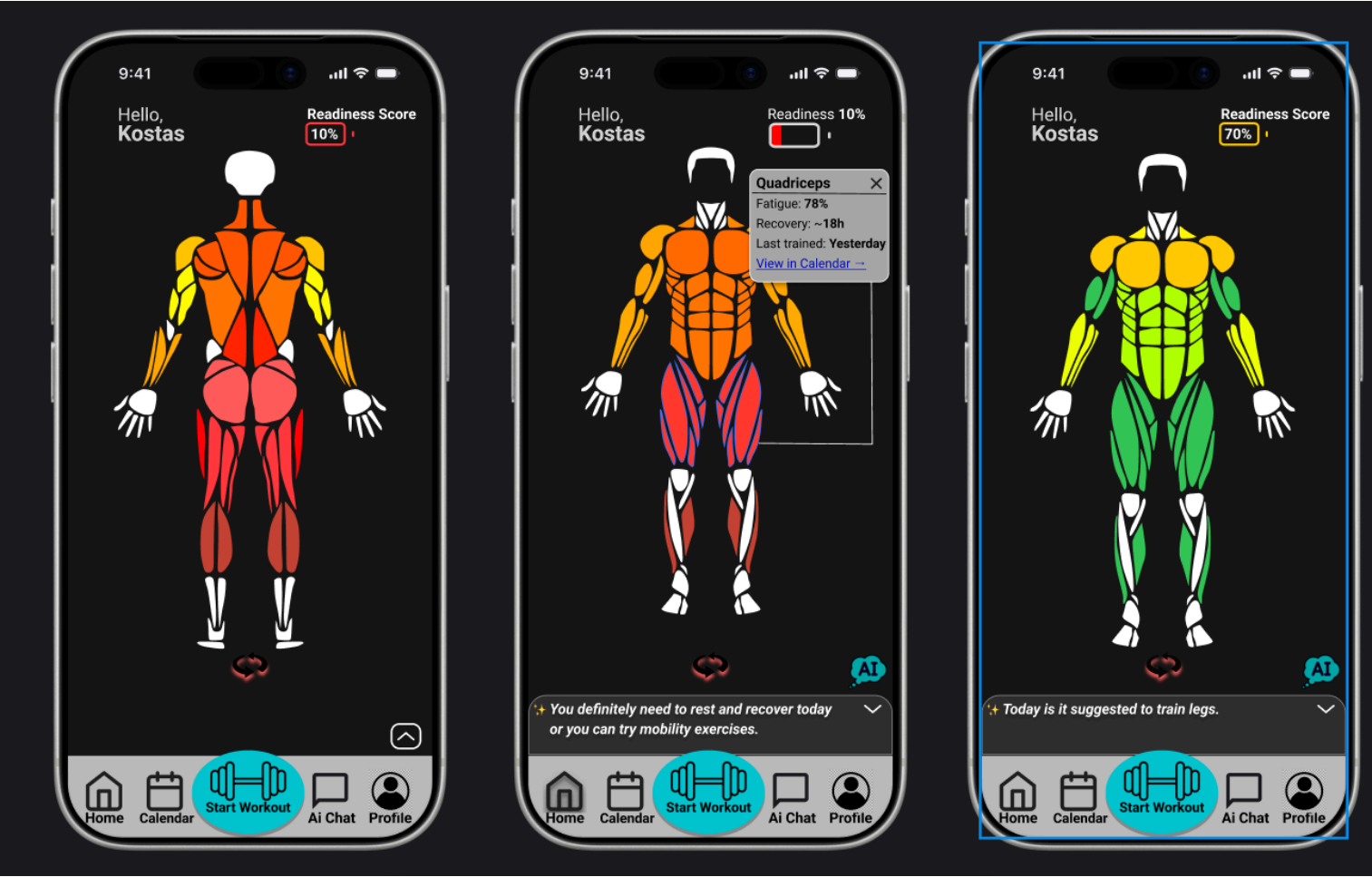






1. User Management Entities

1.1 User

Attributes:

`user_id (PK)`

`email`

`password`

`created_at`

Relationships:

One user has **one user profile**

One user has **one settings record**

One user can have **many workout sessions**

One user can have **many biometric measurements**

One user can have **many fatigue records**

One user can have **many notifications**

One user can have **many chat threads**

1.2 User Profile

Attributes:

user_id (PK, FK)

name

age

height

gender

phone_number

fitness_level

goal

Relationship:

User Profile has a **one-to-one relationship with User**.

1.3 Settings

Attributes:

user_id (PK, FK)

preferred_unit

notification_enable

inactivity_days_threshold

theme

Relationship:

Each user has **one settings configuration**.

2. Exercise Management

2.1 Exercise

Attributes:

exercise_id (PK)
name
modality_id (FK)
description
created_by_user_id (FK)

Exercises can be either:

Global exercises (available to all users)

Custom exercises created by a specific user

2.2 Modality

Examples include:

Weightlifting
Calisthenics
CrossFit
Track & Field
Yoga / Mobility

Attributes:

modality_id (PK)

name

Relationship:

One modality can be associated with **many exercises**.

2.3 Equipment

Examples:

Barbell

Dumbbell

Pull-up bar

Resistance bands

Attributes:

equipment_id (PK)

name

description

2.4 Equipment Exercise

Attributes:

equipment_id (FK)

exercise_id (FK)

qty_optional

notes

3. Muscle Representation

3.1 Muscle

Attributes:

muscle_id (PK)

name

Examples:

Quadriceps

Hamstrings

Chest

Shoulders

3.2 Muscle Path

Attributes:

muscle_path_id (PK)

muscle_id (FK)

svg_path_id

One muscle may have **multiple visual paths** in the body diagram.

3.3 Muscle Exercise

Attributes:

muscle_id (FK)

exercise_id (FK)

impact_factor

The `impact_factor` represents the relative stress an exercise places on a specific muscle group.

4. Workout Logging

4.1 Workout Session

Attributes:

session_id (PK)
user_id (FK)
date_time
duration
avg_rpe
total_volume
weather_condition

Relationship:

One user can have **many workout sessions**.

4.2 Workout Exercise

Attributes:

workout_exercise_id (PK)
session_id (FK)
exercise_id (FK)
order_index
notes

4.3 Workout Set

Attributes:

set_id (PK)

workout_exercise_id (FK)

set_number

5. Set Type Tables

5.1 Set Strength

Attributes:

set_id (PK)

reps

weight

rpe

Used for strength training exercises.

5.2 Set Cardio

Attributes:

set_id (PK)

distance

time

Used for endurance activities.

5.3 Set Calisthenics

Attributes:

set_id (PK)

reps

added_weight

rpe

5.4 Set WOD

Used for CrossFit workouts.

Attributes:

session_id (PK)

wod_type(for_time, amrap, emom)

time_cap

score_value

score_unit

notes

5.5 Set Mobility

Used for yoga and mobility sessions.

Attributes:

session_id (PK)

duration

focus_area

notes

6. Muscle Fatigue Tracking

6.1 Muscle Fatigue Current

Stores the **current fatigue state** of each muscle for a user.

Attributes:

user_id (PK)

muscle_id (PK)

fatigue_level

updated_at

recovery_target_at

6.2 Muscle Fatigue Log

Attributes:

mfl_id (PK)
user_id (FK)
muscle_id (FK)
source
delta
fatigue_level_after
created_at
workout_session_id

7. Biometrics and Health Data

7.1 Biometrics

Attributes:

bio_id (PK)
user_id (FK)
measure
value
source

Examples of measures:

weight

body fat

lean mass

heart rate

HRV

readiness score

7.2 Nutrition Log

Attributes:

nutrition_id (PK)

user_id (FK)

log_date

protein_g

calories

notes

7.3 Sleep Log

Attributes:

sleep_id (PK)

user_id (FK)

sleep_date

duration_min

sleep_score

notes

8. AI Interaction

8.1 Chat Thread

Attributes:

chat_id (PK)

user_id (FK)

created_at

8.2 AI Chat Messages

Attributes:

chat_id (FK)

user_id (FK)

message_text

sender

date_time

9. Notifications

Attributes:

notification_id (PK)

user_id (FK)

type

title

body

status

created_at

sent_at

Notifications may be triggered by:

inactivity

recovery updates

training suggestions

system alerts